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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/273,816	02/06/2024	Clifford P. BRANGWYNNE	Princeton - 73403	1946

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EXAMINER
WHITE, DENNIS MICHAEL

ART UNIT	PAPER NUMBER
1758	

NOTIFICATION DATE	DELIVERY MODE
10/08/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/273,816	Applicant(s) BRANGWYNNE et al.	
	Examiner DENNIS M WHITE	Art Unit 1758	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 8/7/2026.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 31-32,35,37,39,53-57 and 102-111 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 31-32,35,37,39,53-57 and 102-111 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 7/24/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Election/Restrictions

Applicant's election without traverse of claims 31-32, 35, 37, 39, 53-57, and 102-111 in the reply filed on 8/7/2026 is acknowledged.

Claim Rejections - 35 USC § 102

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.

Claim(s) 31-32, 35, 37, 39, 53, 57, 102-110 is/are rejected under 35 U.S.C. 102a1 as being anticipated by Young et al (WO 2020/232416).

Regarding claims 31-32, Young et al teach a method of identifying an inhibitory activity of an exogeneous agent that disrupts binding of a target peptide to a biomolecule (para. 0199: high throughput screening of contacting agents with condensates, Para. 0246: the agent can be an inhibitor), the method comprising:

(i) contacting a cell with said agent, wherein said cell comprises a target molecule that comprises said target peptide, and at least one peptide selected from a light-sensitive receptor and a chemical-sensitive receptor (Para 0058, Fig. 24: estrogen receptor/MED1 droplets with fluorescent peptide, the ER undergoes a conformational change to allow MED1 LXXLL domain interaction);

(ii) exposing said cell to at least one predetermined wavelength of light, a chemical to which said chemical-sensitive receptor is sensitive, or a combination thereof (Para. 0005, 0167; p. 120, claim 3: detected using spectrophotometry); and

(iii) detecting membrane-less bodies comprising said target molecule, wherein said target molecule is incorporated into said membrane-less bodies when said target peptide is released from said binding to said biomolecule and undergoes phase separation at a sub-region of said cell that is separate and distinct from a sub-region where said target peptide binds to said biomolecule (Para. 0059: LXXLL peptide added to ER/MED1 droplets, peptide incorporated into MED1 resulting in partitioning of ER into MED1; LXXLL peptide reads on "inhibitor target peptide"); wherein said detecting comprises measuring a change in an intensity and a size of said membrane-less bodies wherein a bigger change in said intensity and said size of said membrane-less bodies relative to a change in an intensity and a size of membrane-less bodies in the absence of said agent indicates that said agent disrupts binding of said target peptide to said biomolecule (Para. 0172: signal intensity of the agent quantifies the condensate partition coefficient based on the labeled component of the condensate).

Specifically regarding claim 32, Young et al teach (i) contacting a cell with said agent, wherein said cell comprises (a) a target molecule that comprises said target peptide and a peptide selected from a group consisting of a light-sensitive receptor, a chemical-sensitive receptor, a light-sensitive oligomerization protein, and a non-light sensitive dimerization module, or a cognate partner thereof (Para. 0130: cognate ligand); and (b) a second molecule comprising the other of said peptide, or cognate partner thereof, fused to a full length or truncated low complexity protein region or a full length or

truncated intrinsically-disordered protein region (IDR) (Para. 0130: functional fragment comprising an IDR and LXXLL motif);

Regarding claim 35, Young et al teach the target molecule further comprises a fluorophore fused between said target peptide and said peptide. (para. 0094: ER fused to GFP and MED1)

Regarding claim 37, Young et al teach said detecting membrane-less bodies is measuring fluorescence, wherein said measuring fluorescence comprises detecting an intensity of fluorescence, quantifying the number of said membrane-less bodies, or a combination thereof (Para. 0172: fluorescent signal intensity).

Regarding claim 39, Young et al teach the membrane-less bodies are nucleoli, Cajal bodies, Stress granule, P-bodies, spliceosomes, or a combination thereof. (Para. 0121, 0140: nucleoli)

Regarding claim 53, Young et al the IDR is at least a portion of FUS, a portion of Ddx4, a portion of hnRNPA1, a portion of BRD4, a portion of TAF15, a portion of SRSF2 IDR, a portion of SART1, a portion of HSF1, a portion of RNPS1, or a combination (Para. 0191: SRSF2).

Regarding claim 57, Young et al teach said fluorophore is mCherry or green fluorescent protein (GFP) . (para. 0094: ER fused to GFP and MED1)

Regarding claim 102, Young et al teach biomolecule is a protein, a protein domain, DNA, RNA, or a complex comprising a protein, DNA, RNA, or a combination thereof (Para. 0246: transcription condensate include protein).

Regarding claims 103-105, Young et al teach said agent is a small molecule compound; said agent is selected from the group consisting of a nucleic acid, a polynucleotide, a polypeptide, or a combination thereof; wherein said polypeptide is an antibody or a functional fragment thereof. (Para. 0345: MED1 antibody)

Regarding claims 106-110, Young et al teach the polynucleotide is an aptamer, a ribonucleic acid (RNA)-based compound, an antisense nucleic acid, a peptide nucleic acid (PNA), or an combination

thereof; said RNA-based compound is a small interfering RNA, a microRNA, a small hairpin RNA, or a combination thereof; said polynucleotide is a CRISPR guide RNA; said CRISPR guide RNA is encoded by a viral vector; said CRISPR guide RNA is introduced in combination with Cas9 or a vector comprising a nucleotide sequence encoding said Cas9 (Para. 0318: CRISPR/Cas9).

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

This application currently names joint inventors. In considering patentability of the claims the examiner presumes that the subject matter of the various claims was commonly owned as of the effective filing date of the claimed invention(s) absent any evidence to the contrary. Applicant is advised of the obligation under 37 CFR 1.56 to point out the inventor and effective filing dates of each claim that was not commonly owned as of the effective filing date of the later invention in order for the examiner to consider the applicability of 35 U.S.C. 102(b)(2)(C) for any potential 35 U.S.C. 102(a)(2) prior art against the later invention.

Claim(s) 54-56 is/are rejected under 35 U.S.C. 103 as being unpatentable over Young in view of Brangwynne et al (US 20180251497 A1)

Regarding claim 54-56, Young et al teach the IDR is at least a portion of SRSF2 IDR, Young et al is silent to said portion is portion of FUS is an N-terminal IDR of FUS (FUSn); said light-sensitive receptor is cryptochrome 2 (Cry2), a photolyase homology region of Cry2, phytochrome B (PhyB), PIF, a light oxygen voltage sensing (LOV) domain, Dronpa, or an improved light-induced dimer (iLID); and light-sensitive receptor is ssrA and said cognate partner is sspB.

Brangwynne et al teach a method controlling protein assemblies including droplets for assembling membraneless bodies where the IDR cognate partner (40) fused to an IDR and light sensitive receptor component (20), which may include but is not limited to ssrB, Zdk, CIB, or PIF for LOV2-ssrA, LOV2, Cry2, or PhyB respectively. In preferred embodiments, the second protein construct comprises an IDP (60), which include but not limited to full length or truncated forms of FUS, DDX4, and hnRNPA1. In some embodiments, the IDR comprises amino acids 1-214 of FUS, 1-236 of DDX4, or 186-320 of HNRNPA1. The fluorophore (50) can comprise any appropriate fluorescent tag, such as mCherry, although the use of other fluorescent proteins is also envisioned, including but not limited to GFP variants.(Para. 0027-0028). It is advantageous to provide the above light sensitive receptors to control activation and deactivation with specificity (Para. 0044-0046). Combining prior art elements according to known methods to yield predictable results is known. Therefore it would have been obvious to one of ordinary skill in the art to combine the IDR and light sensitive receptors of Brangwynne to the method of Young to provide the above advantage of controlling activation and deactivation with specificity.

Claim(s) 111 is/are rejected under 35 U.S.C. 103 as being unpatentable over Young et al in view of Larson et al, "CRISPR interference (CRISPRi) for sequence-specific control of gene expression" Nature Protocols volume 8, pages2180–2196 (2013).

Regarding claim 111, Young et al teach Cas9, but is silent to wherein said Cas9 is enzymatically dead Cas9 (dCas9). Larson et al teach CRISPR Cas9 using the catalytically inactive Cas9 (dCas9) for halts transcription but has the advantage of not causing the endonuclease to cleave, but retains the ability to guide RNA directed DNA binding at targeted genomic loci. (Larson: p. 2181 col. 1-2). Simple substitution of one known element for another to obtain predictable results is held to be obvious. Therefore, it would have been obvious to one of ordinary skill in the art to substitute the CRISPR/Cas9 of Young et al with the CRISPR/dCas9 to provide the above advantage of preventing the endonuclease to cleave, but retains the ability to guide RNA directed DNA binding at targeted genomic loci.

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to DENNIS MICHAEL WHITE whose telephone number is (571)270-3747. The examiner can normally be reached M-F 8:30am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Maris R. Kessel can be reached at (571) 270-7698. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Dennis White/
Primary Examiner, Art Unit 1758